

kmhelpers: A Python toolkit for automated management of k -mer indices

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Summary

Large-scale genomic sequence search has become a central challenge in modern bioinformatics. k -mer indices built on Bloom filters, such as those produced by `kmindex` (Téo Lemané & others, 2024), allow efficient querying of large sequence collections. However, building and managing such indices in practice involves a complex chain of steps: enumerating sample files, counting k -mers, selecting appropriate Bloom-filter parameters, generating index definitions, orchestrating build jobs, querying indices, and compressing outputs for long-term storage. Each step requires expertise, and errors at any stage can invalidate downstream results or waste significant compute time.

`kmhelpers` is an open-source Python toolkit that automates this entire workflow. It provides both a command-line interface (CLI) and a Python API, covering every stage of the k -mer index lifecycle:

- sample discovery and k -mer counting (`list`),
- Bloom-filter span profiling and grouping (`profile`),
- index definition generation (`compose`),
- upfront validation of sample files, available disk space and memory with generation of ready-to-execute pipeline scripts (`plan`),
- index building from definition files (`apply`),
- sequence querying (`query`),
- ZSTD-based index (block) compression (`compress`) (?),
- registry management (`registry`).

Multi-step workflows can be described as declarative YAML pipelines (`pipeline`) and executed in a single command.

Statement of Need

k -mer based sequence databases built with tools like `kmindex` and `kmtricks` (Thibault Lemané et al., 2022) are used to address large-scale questions in genomics, metagenomics, and population studies. However, the gap between having the raw indexing software and running a reproducible, end-to-end indexing workflow is significant. Researchers face recurring pain points:

Parameter selection. Bloom filter-based indices require choosing a filter size (or *span*) per sample. Choosing this parameter incorrectly leads to either storage consumption or unacceptably high false-positive rates. `kmhelpers` automates this with its `profile` command, which reads k -mer count distributions produced by `ntCard` (Mohamadi et al., 2017) and assigns each sample to the smallest span that keeps the false-positive rate below a user-defined threshold.

Workflow orchestration. Building a large index may involve hundreds to millions of samples

spread across multiple sub-indices. Managing these build jobs manually is error-prone. kmhelpers introduces a declarative index definition format (YAML) and two complementary commands: `plan`, which validates sample files, available disk space and memory upfront and emits ready-to-execute pipeline scripts; and `apply`, which reads the definition files and runs the build, with support for span-level and name-level filtering.

Together, these features make k -mer indexing workflows accessible to researchers who are not experts in the underlying data structures, while remaining flexible enough for large-scale production use.

State of the Field

Several tools address large-scale sequence search using k -mer based data structures. BIGSI (Bradley et al., 2019) and COBS (Bingmann et al., 2019) use compressed Bloom filter matrices to answer presence/absence queries across collections of sequencing experiments. HowDeSBT (Harris & Medvedev, 2020) and Mantis (Pandey et al., 2018) use sequence Bloom trees and counting quotient filters, respectively, to support similar queries. kmindex, which kmhelpers wraps, builds on the kmtricks counting pipeline and is designed for efficient querying of large sequence collections.

kmhelpers does not compete with any of these approaches at the algorithmic level. Its contribution is orthogonal: it provides the workflow automation, parameter optimisation, and index lifecycle management that are necessary to deploy kmindex-based databases in practice, but are absent from the core tool. Similar automation layers exist for other complex bioinformatics pipelines (e.g., Snakemake (Köster & Rahmann, 2012) workflows wrapping alignment or assembly tools), but no dedicated automation layer existed for kmindex prior to kmhelpers.

Design and Implementation

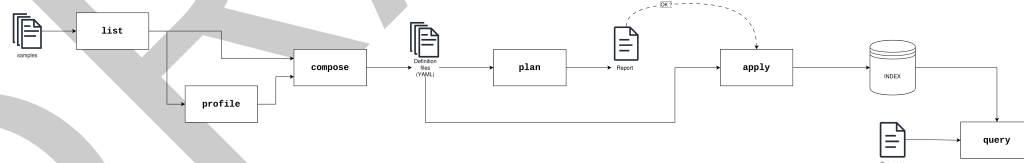


Figure 1: Overview of the kmhelpers workflow. `list` enumerates sample files and counts k -mers; `profile` analyses the count distributions to assign each sample to a Bloom-filter span and recommend index groupings. Both outputs feed into `compose`, which generates YAML index definition files. `plan` then validates sample paths, available disk space and memory, and emits a ready-to-execute pipeline script; the resulting report is reviewed before committing to the build. `apply` reads the definition files and invokes kmindex to construct the index. Finally, `query` searches the built index against user-provided sequences and returns ranked results.

[[TODO]]

kmhelpers is implemented in Python (3.8) and distributed via Conda with automatic installation of its bioinformatics dependencies (kmindex, ntCard). The CLI is built with Click (Ronacher & Click Contributors, 2023) and the package exposes a public Python API covering all CLI functionality.

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AI Usage Disclosure

AI assistance was used for constrained tasks (drafting, editing, code suggestions) under strict human review at every stage. AI provider used: Claude (Anthropic, 2025).

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